

# Solange Daniela Correa Holguin

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## SOFTWARE & CLOUD ENGINEERING

SOFTWARE ENGINEERING | CLOUD COMPUTING (AWS & AZURE) | FULL-STACK DEVELOPMENT | DATABASE SYSTEMS | AGILE & CI/CD | NETWORKING FUNDAMENTALS

### TECHNICAL SKILLS

- **Programming:** Python | C++ | C# | JavaScript | SQL | Java | Bash | PowerShell | HTML | CSS | Linux
- **Frameworks & Tools:** React | Node.js | .NET | WPF | Tkinter | Pandas | XAML | Excel Interop
- **Databases:** MySQL | MongoDB | DynamoDB | SQL Server | ER diagrams | CRUD | Triggers, Normalization
- **Cloud:** AWS (EC2, Lambda, API Gateway, DynamoDB, S3, IAM, Load Balancer) | Azure (VMs, SQL, VPN, SSH tunneling, scaling, performance, monitoring)
- **Networking & Systems:** TCP/IP | UNIX/Linux | Routing & Switching (BGP, OSPF) | Network Troubleshooting | Packet Analysis (Wireshark)
- **Practices:** Agile/Scrum | CI/CD pipelines | SDLC | OOP | MVC | Infrastructure as Code (IaC) | Debugging | Code Reviews | System Monitoring & Troubleshooting
- **Other Tools:** Git | GitHub | MySQL Workbench | Visual Studio | VS Code
- **Soft Skills:** Strong Communication | Team Collaboration | Problem Solving | Time Management | Adaptability | Bilingual (English, Spanish)

### WORK EXPERIENCE

Winthrop University - Rock Hill, SC

Aug 2024 – May 2025

#### Software Engineer | Dowdy Excel Project (Capstone Project - Apprenticeship)

Designed and delivered a **reporting and export system** for a nonprofit client, hosted on **Microsoft Azure**. Built a **Python/Tkinter UI** with a **MySQL backend** to automate data reporting and enable exports into **Excel, Word, and PDF**.

- Built **dynamic filters** (Program Title, Main Category, Subcategory, Date Range) enabling stakeholders to isolate and analyze specific data segments.
- Applied **SDLC principles** (requirements gathering, design, implementation, testing, and deployment) to deliver a secure and maintainable solution.
- Implemented **data validation, SQL injection protection, and error handling** to enhance reporting **security and compliance reliability**.
- Developed export functionality using **Pandas** and **fpdf**, ensuring accurate data transfer into Excel, Word, and PDF reports.
- Configured **Azure-hosted MySQL server** with **SSH tunneling** and migrated production data with schema refactoring.
- Conducted **unit/integration testing, debugging, code reviews**, ensuring functionality and maintainability across the system lifecycle.
- Delivered client demos, Agile sprint updates, and comprehensive documentation to support long-term adoption and maintenance.

Winthrop University - Rock Hill, SC

Jan 2025 – May 2025

#### Cloud Developer | (AWS Cloud Projects - Apprenticeship)

Designed and deployed cloud-based applications on **Amazon Web Services (AWS)**, focusing on scalability, automation, and secure infrastructure management.

- Deployed a **scalable, load-balanced web application** using **Amazon EC2, Elastic Load Balancer (ALB)**, and **custom security groups** to simulate high availability in production.
- Built a **serverless backend** with **AWS Lambda, API Gateway, and DynamoDB**, supporting **RESTful API** operations without server management.
- Configured and mounted **Amazon EFS** across **EC2 instances**, integrating with **Lambda functions** for shared storage and automated data processing.
- Implemented **least-privilege IAM roles** with custom **JSON policies**; **tested and debugged access controls** across multiple environments (FullAccess, ReadOnlyAccess, Admin).
- Applied **CI/CD principles** with **automated testing and monitoring** using **GitHub workflows** to streamline deployments and ensure reliability.

Winthrop University - Rock Hill, SC

Aug 2024 – Dec 2024

#### Network Programmer | Reliable Data Transfer (RDT) Protocol Simulator Project

Developed a Python implementation of **rdt3.0 (Alternating Bit Protocol)** to ensure in order, error-free delivery over an unreliable channel.

- Designed **sender (EntityA)** and **receiver (EntityB)** with **checksums, ACK/NACK validation, retransmission timers, and sequence control**.
- Simulated **packet loss, corruption, and network delay**, testing and debugging reliability through automated experiments and performance summaries.
- Applied **TCP/IP reliability principles (flow control, retransmission logic)** within a **UNIX-like simulation environment**, strengthening **transport layer fundamentals**.

Carolina IT Solutions - Charlotte, NC

Sept 2024 - Sept 2025

#### IT Support Technician

Delivered Tier 1 technical support, improving issue resolution time by 25% and increasing user satisfaction.

- Configured **Azure VPN gateways** and deployed **Office 365 accounts**, strengthening secure remote access and productivity.
- Monitored and maintained **Hyper-V VMs** supporting business apps, reducing downtime by 30%.
- Automated repetitive tasks with **PowerShell scripts**, documented issues in a ticketing system, and enhanced reporting accuracy.

|                                                                                                                                                        |                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Snorkel AI</b> - Remote                                                                                                                             | Nov 2025 - Present |
| <b>AI Task Developer (Freelance Project)</b>                                                                                                           |                    |
| <ul style="list-style-type: none"><li>Freelance project onboarding</li><li>Training / evaluation tasks</li><li>Exploring areas to contribute</li></ul> |                    |

EDUCATION

|                                                                                                                                                                                     |                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Winthrop University</b> - Rock Hill, SC                                                                                                                                          | Jan 2024 - May 2025 |
| Bachelor of Science - Computer Science, Minor in Mathematics                                                                                                                        | GPA: 3.67           |
| Inducted Honors: Upsilon Pi Epsilon (UPE) International Honor Society in Computing                                                                                                  |                     |
| Relevant Coursework: Software Engineering, Networking, Operating Systems, Cloud Computing, C#, C++, Python, Java, JavaScript                                                        |                     |
| <b>UNIVERSITY OF NORTH CAROLINA</b> - Charlotte, NC                                                                                                                                 | May 2024 - Aug 2024 |
| Summer 2024 Coursework in Computer Science                                                                                                                                          | GPA: 4.00           |
| Relevant Coursework: Introduction to Computer Science II, Intro to Computer Systems, Data Structures & Algorithms                                                                   |                     |
| <b>ESPOL (Higher Polytechnic School of the Litoral)</b> – Guayaquil, Ecuador                                                                                                        | Jan 2015 – Jan 2018 |
| Bachelor of Telecommunication and Electronic Engineering (Incomplete – 3 years equivalent per WES)                                                                                  | GPA: 3.70           |
| WES evaluation: Equivalent to three years of U.S. regionally accredited undergraduate study.                                                                                        |                     |
| Relevant Coursework: Electrical Network Analysis I & II, Telecommunications Fundamentals, Programming Fundamentals, Programming for Telecommunications, Digital Collaboration Tools |                     |

CERTIFICATIONS

- AWS Certified Cloud Practitioner – In Progress (Expected Dec 2025)